Analog to Digital Conversion

ECE 2564, Embedded Systems

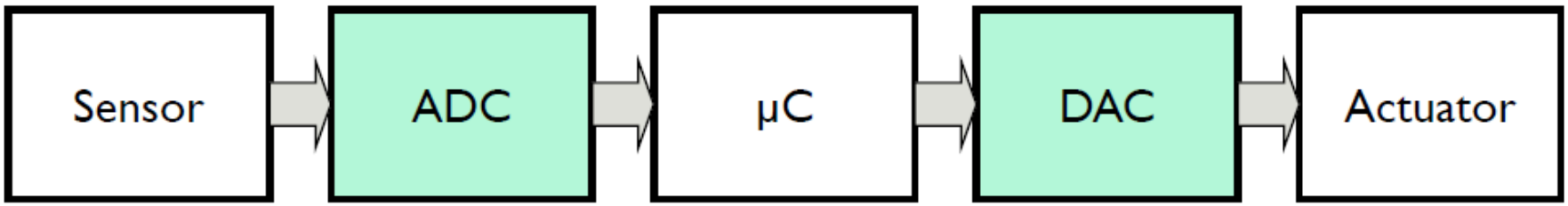
Fall 2025

Acronyms

- Analog-to-Digital
  - ADC
  - A/D
- Digital-to-Analog
  - DAC
  - D/A

Motivation

- Nature provides us with many physical quantities that vary *continuously* with time: current, voltage, temperature, pressure, humidity, etc.
- Most digital systems require discrete time steps and discrete signal values
- A CPU needs to interact with the physical world
  - An ADC is used to convert the analog sensor input to a digital value
  - A DAC is used to convert a digital value to an analog signal that can be used by an actuator

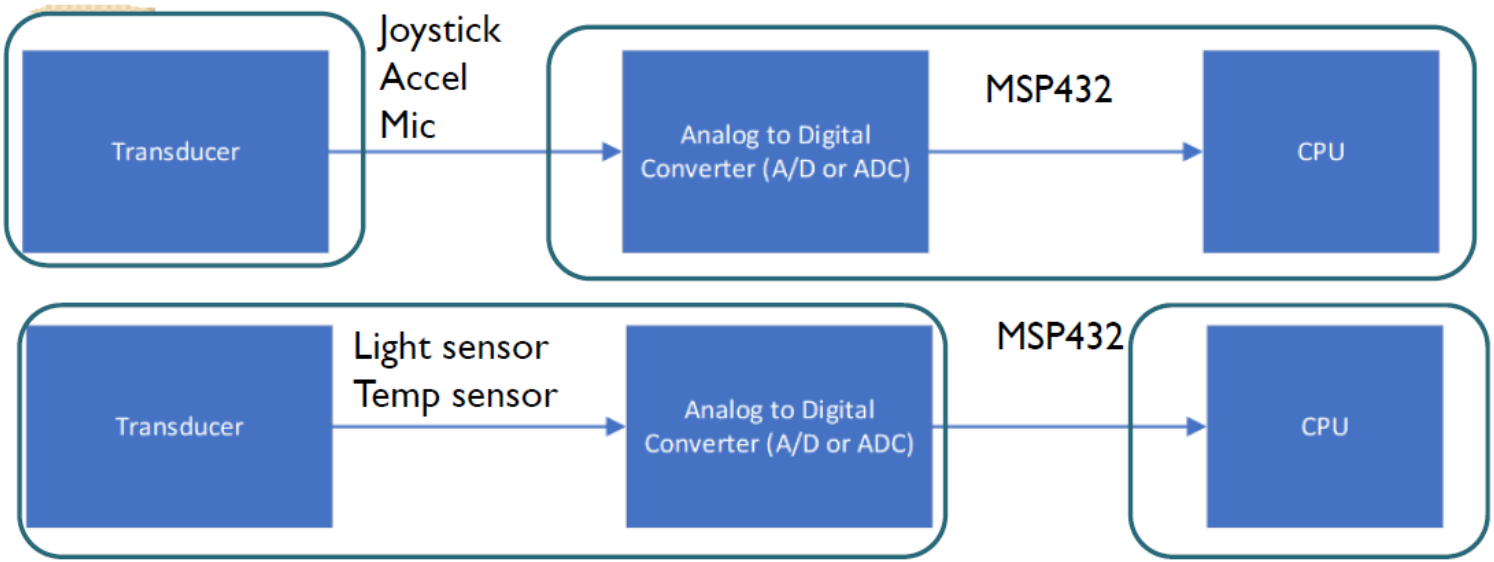


Transducers

- Sensors and actuators are *transducers*
  - A transducer converts energy from one form to another
  - Example: a photodiode produces an electrical current as a function of the photons striking the photodiode device
- Consider the following transducers:
  - Light sensor — converts light to electricity
  - Pressure sensor – converts pressure to electricity
  - LED — converts electricity to light
  - Microphone — converts sound to electricity
  - Speaker — converts electricity to sound
  - Joystick —converts mechanical position to electricity

Examples of Inputs and Sensors

- Position — Potentiometer
- Direction — Gyroscope
- Temperature — Thermistor
- Force, pressure — Accelerometer, strain gauge
- Sound — Microphone
- Light intensity — Photodiode, camera
- Radio signals — antenna



Examples of Outputs and Actuators

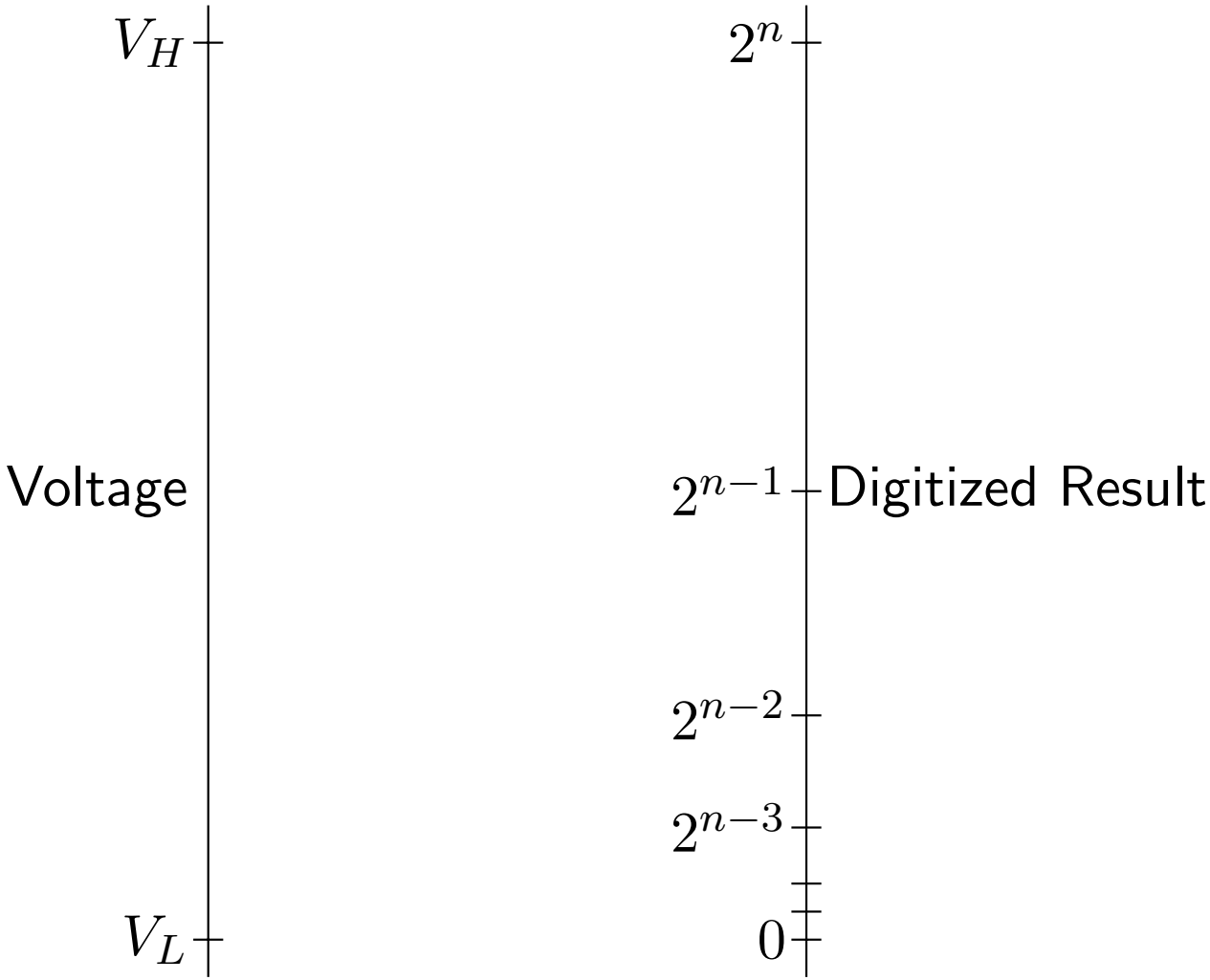
- Position, orientation — Motor
- Sound — Acoustic Speaker
- Electrical current — Relay

- We will discuss more about outputs and actuators when we cover DAC

Analog-to-Digital Conversion

- Goal: convert an analog signal to an unsigned binary number
  - The input signal is usually a voltage in a common input range, such as 0 to V_{CC} or -12 to $12\,V$
- Analog-to-digital conversion normally uses a linear mapping
- The key concerns when performing the conversion are
  - *resolution*, *accuracy* and *speed*
- The sampled voltage should be mapped to the closest quantized binary value

The Fundamental Idea



Analog-to-Digital Conversion

- The conversion is described here:

$$D = \left\lfloor \frac{V_{\text{in}} - V_L}{V_H - V_L} \cdot 2^n \right\rfloor$$

- V_{in} is the *analog input voltage* to be converted
- D is the *digitized result* of the conversion (unsigned decimal equivalent)
- V_H and V_L are the *high reference* and *low reference* voltages
- n is the *number of bits* used to represent the output of the converter
- *Span* is the total analog voltage range of the input, $V_H - V_L$
  - Sometimes called the Effective Full-Scale Range, EFSR
- Voltage *resolution*: the span divided by 2^n
  - the voltage step corresponding to the least significant bit (LSB)

Example Output from an ADC

- $n = 14$ bits
- $2^n = 16384$

14-bit Output Code	Decimal equivalent
11 1111 1111 1111	16383
11 1111 1111 1110	16382
11 1111 1111 1101	16381
10 0000 0000 0001	8193
10 0000 0000 0000	8192
01 1111 1111 1111	8191
00 0000 0000 0010	2
00 0000 0000 0001	1
00 0000 0000 0000	0

ADC Example

- $V_H = 4\,\mathrm{V}$
- $V_L = 0\,\mathrm{V}$
- $n = 10$

- Suppose $V_{\mathsf{in}} = 2.02\,\mathrm{V}$

$$
\begin{aligned}
D = \left\lfloor \frac{V_{\mathsf{in}} - V_L}{V_H - V_L} \cdot 2^n \right\rfloor &= \left\lfloor \frac{2.02 - 0}{4 - 0} \cdot 2^{10} \right\rfloor \\
&= \lfloor 0.505 \cdot 1024 \rfloor = \lfloor 517.12 \rfloor = 517 \\
&= 1000000101_2
\end{aligned}
$$

Digital-to-Analog Conversion

- The conversion is described here:

$$V_{\text{out}} = \frac{D}{2^n}(V_H - V_L) + V_L$$

- D is an unsigned digital value to be converted to analog form
- V_{out} is the *analog output voltage* result of the conversion
- V_H and V_L are the *high reference* and *low reference* voltages
- n is the number of bits that are input to the converter
- Span is the total analog voltage range of the output, $V_H - V_L$
  - Sometimes called the Effective Full-Scale Range, EFSR
- Voltage *resolution*: the span divided by 2^n
  - the voltage step corresponding to the least significant bit (LSB)

ADC and DAC: Common Concepts

What are the primary concerns in achieving the conversion?

- Resolution
  - Voltage resolution: How close can two voltages be and still be distinguishable from each other?
  - Digital resolution: How many bits are used to represent digital values? (How many quantization levels?)
- Accuracy
  - Inputs: How close is a measure value to the true state of nature?
  - Outputs: How close is an actual output value to the desired theoretical value?
- Speed: How fast cand the converter produce the output?
- Precision How repeatable are the conversions?

Resolution

Number of quantization levels, $Q = 2^n$

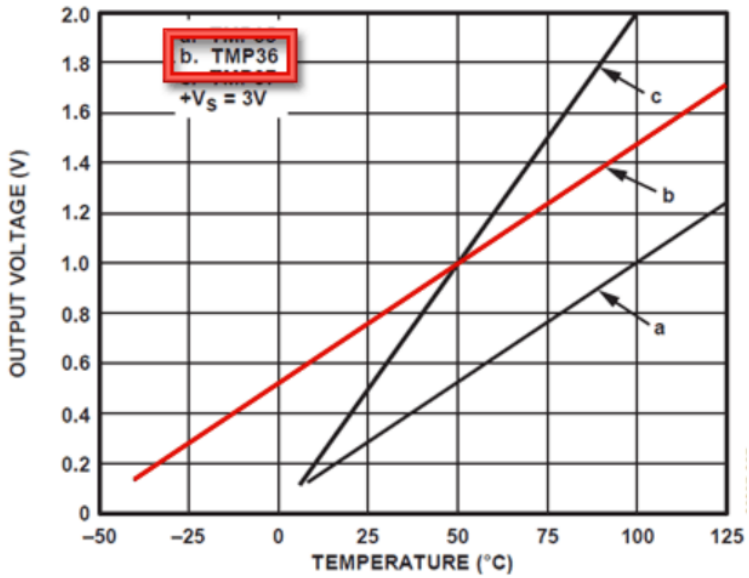
- It determines how closely the digitized signal can mimic the original input signal
- It has a large effect on the complexity and the price of the ADC

ADC and DAC: Common Concepts

- The goal of a typical A/D conversion is to map an analog voltage to an unsigned binary number
- The goal of a typical D/A conversion is to map an unsigned binary number to an analog voltage
- The mapping is usually *linear*, which allows us to treat the mapping like solving proportions.

Example Linear Input-Output Relationship for Temperature Sensor [1]

- Linear input-output relationship over a range
- Linearity simplifies conversion



Output voltage versus temperature

[1] Adafruit:https://learn.adafruit.com/tmp36-temperature-sensor

ADC Key Steps

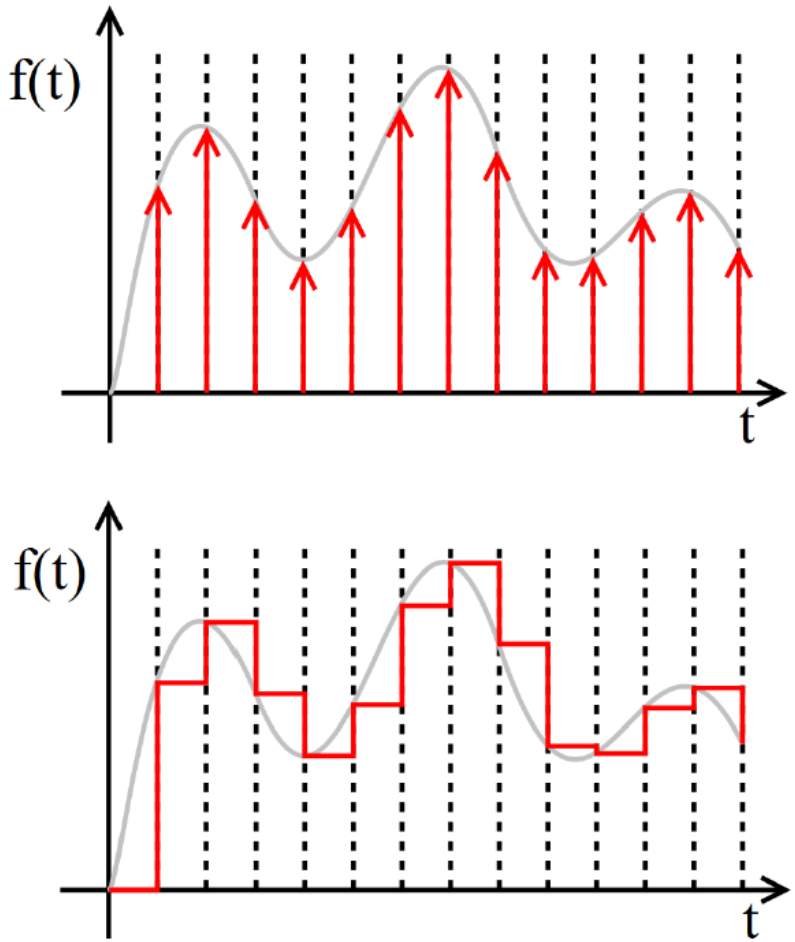
- The transformation of an analog electric signal, voltage or current, to digital has two key steps:
  - Sample-and-hold (S/H)
    - We break the continuity of time
    - This leads to the quantization of time
  - A/D conversion
    - We break the continuity of the analog signal value
    - This leads to the quantization of the electric signal

Sample-and-Hold

- Goal: provide a *steady* voltage level to the ADC during conversion
- Operating mode
  - *Sample mode*: "Latch" the voltage value
  - *Hold Mode*: The sampled input is held constant for the conversion

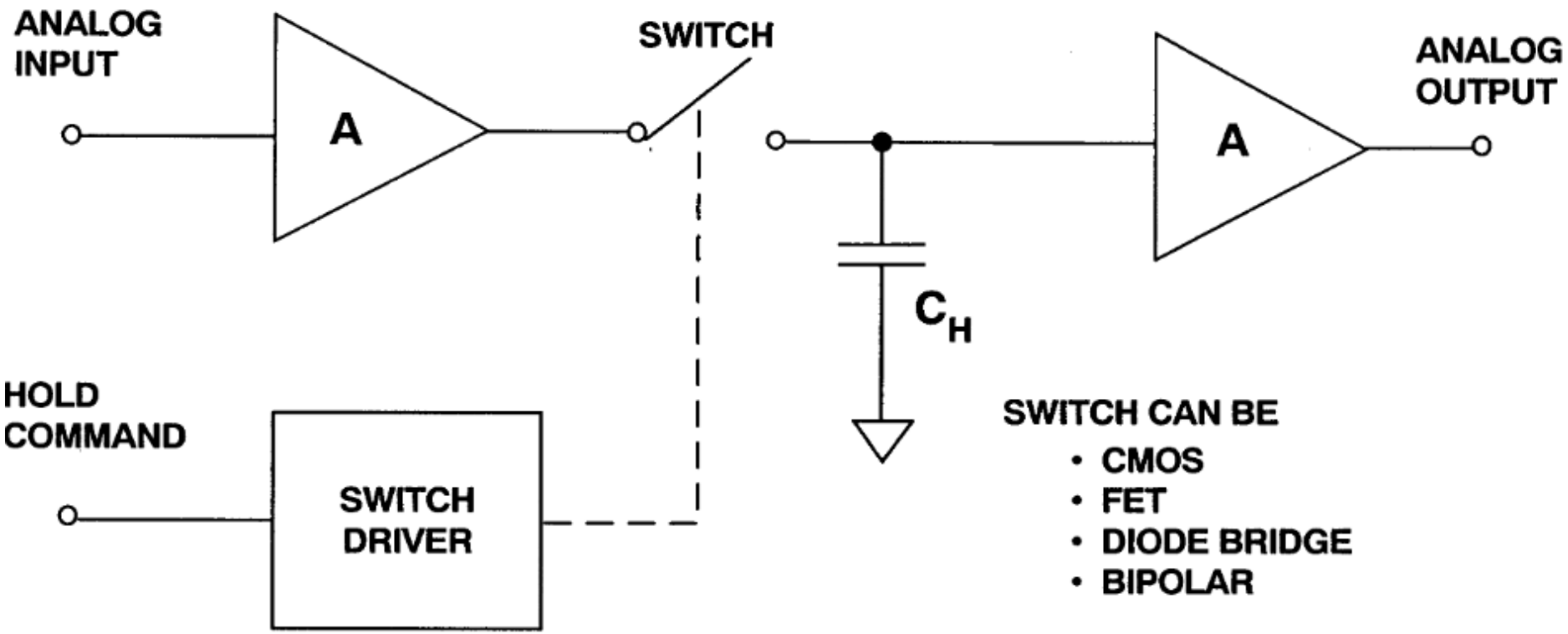
Sample-and-Hold Key Functions

- Purpose: to hold a voltage level steady as and A/D conversion is performed
- Two phases:
  - Sample or "track"
  - Hold or "convert"



Sampling and holding the input signal

Conceptual Sample-and-Hold Circuit [2]

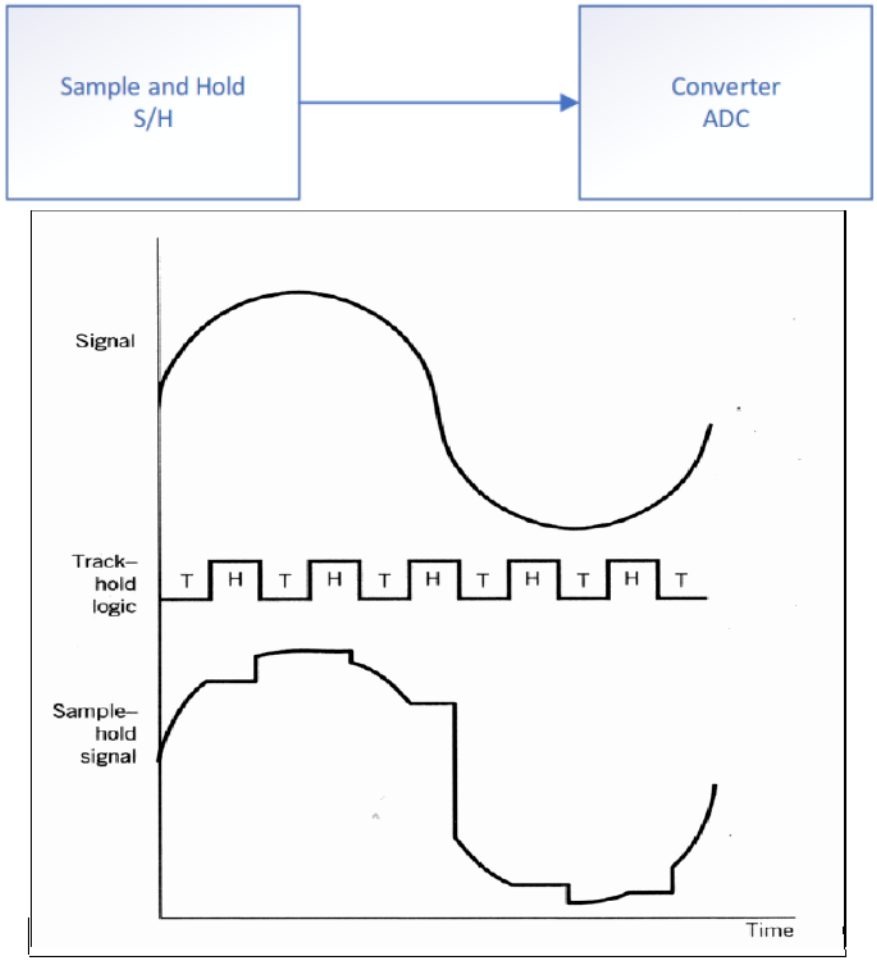


Basic Sample-and-Hold Circuit

[2] Source: https://www.analog.com/media/cn/training-seminars/tutorials/mt-090.pdf

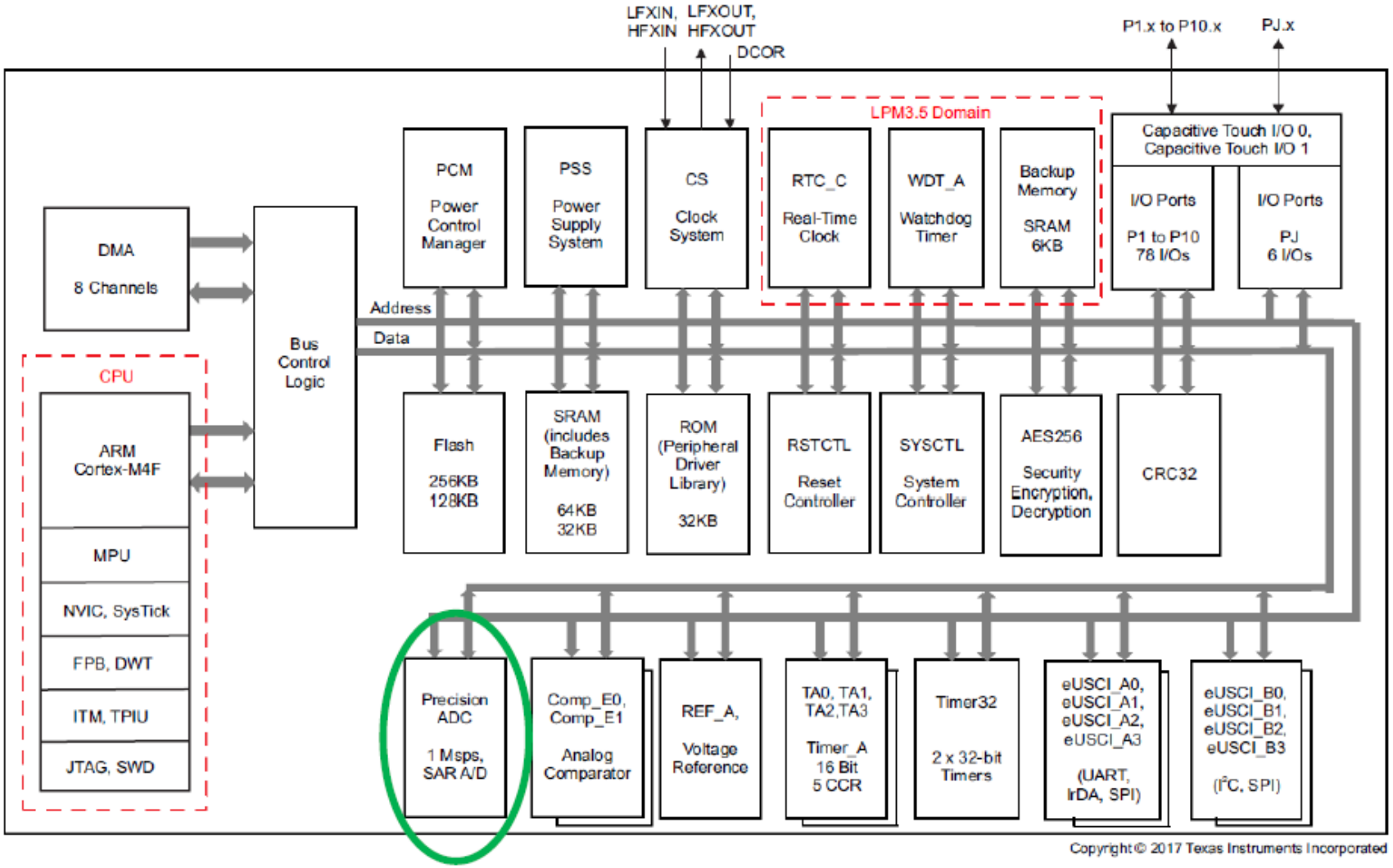
Sample and Hold

- Step1: Sample and Hold
- Step2: Convert to digital



Tracking and Holding

MSP432 Block Diagram



Terminology

- Precision ADC module
  - As used here, this is a marketing term
- 1 Msps maximum conversion rate
  - One million samples per second
- SAR
  - Successive Approximations Register

ADC on the MSP432

- Many analog inputs are possible
  - Up to 32 external input channels; MSP432P4 package supports 24
  - Several internal input channels, including temperature and battery
- Several alternatives for reference voltages
- Six possible clock sources for the ADC
- Conversions can be initiated by software or by timers
- Several conversion sequence modes:
  - Single-channel, single-conversion
  - Sequence-of-channels
  - Repeat-single-channel
  - Repeat-sequence-of-channels
- Conversion result output buffer of 32 registers

Programming the ADC on the MSP432

- Many options are available
- Make use of the User Manual to see the ports/pins
- We will not cover every device in detail
  - We will cover the joystick
  - We will learn to initialize the parameters in the ADC by looking the the reference manual and DriverLib

Summary

- Motivation
- Basic idea
- Transducers, sensors, actuators
- Linear mapping of analog voltage to digital values
- Lots of terminology

- Coming attractions: ADC on the MSP432